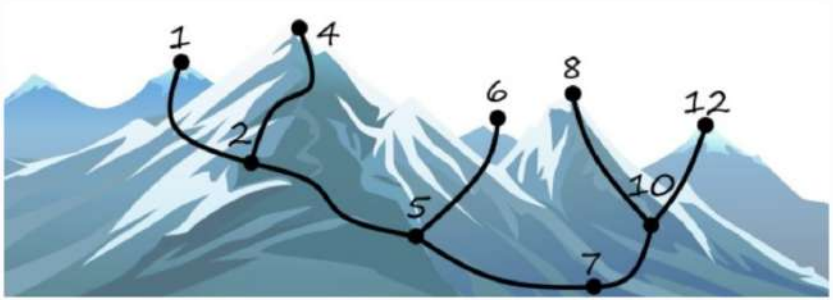
Problem A: 111284. Mountains

You are going to hike in the mountains. You have written directions (left or right) on a piece of paper, in which direction you need to turn on each of the branches of the path, following which you can reach the peak. In the mountains there are several peaks and you have recorded the path to each peak. Since you are a guide, you need to check the day before the hike which of the recorded paths is available. All paths start at one point at the foothills.

The path is presented in the form of "$RRLLRLR$", which means to reach the peak you need to turn right at the beginning, then right again, then left, left, right, left, right. Peak located after last turn and it is possible that it does not exist. You are given all the paths in the mountains that are available in the form of a Binary Search Tree and the paths to the peaks that are written on a piece of paper. You need to tell which of the paths written on the piece of paper is available.



Input format

The first line contains two integers N and M ($1 \le N \le 10^5, 1 \le M \le 2 \cdot 10^4$) — the number of nodes of the tree and the number of paths written on the piece of paper.

The second line contains N integers $a_1, a_2, \ldots, a_N$ ($1 \le a_i \le 10^9$) — the values that are inserted one by one, in this order, into an initially empty binary search tree. A value equal to the value of the current node goes to the left subtree.

Each of the next M lines contains one path p_i ($2 \le |p_i| \le 100$) — a string consisting only of the characters 'L' and 'R'. The total length of all paths does not exceed $2 \cdot 10^6$.

Output format

Print M lines, in the same order as the paths are given. The ith line should contain "YES" if the path p_i leads to an existing node of the tree, and "NO" otherwise.

Examples

Input format

The first line contains two integers N and M ($1 \le N \le 10^5, 1 \le M \le 2 \cdot 10^4$) — the number of nodes of the tree and the number of paths written on the piece of paper.

The second line contains N integers $a_1, a_2, \ldots, a_N$ ($1 \le a_i \le 10^9$) — the values that are inserted one by one, in this order, into an initially empty binary search tree. A value equal to the value of the current node goes to the left subtree.

Each of the next M lines contains one path p_i ($2 \le |p_i| \le 100$) — a string consisting only of the characters 'L' and 'R'. The total length of all paths does not exceed $2 \cdot 10^6$.

Output format

Print M lines, in the same order as the paths are given. The ith line should contain $"YES"$ if the path p_i leads to an existing node of the tree, and $"NO"$ otherwise.

Examples

Input

```
9 4
7 10 12 8 5 6 2 1 4
LLL
LRR
RL
RR
```

Output

```
YES
NO
YES
YES
```

Submit a solution for B-105587. Get subtree

Time limit:	1 s
Real time limit:	5 s
Memory limit:	256M

Problem B: 105587. Get subtree

You are given Binary Search Tree. Your task is to calculate the size of the subtree of the node X.

Remember, that the subtree of node X is the set of all nodes whose ancestor is node X, including it. The size of the subtree is the size of such set.

Input format

The first line of the input contains an integer N — number of nodes in Binary Search Tree ($1 \leq N \leq 10^3$).

The second line contains N pairwise distinct integers a_i ($1 \leq a_i \leq 10^9$) — values of nodes in order of insertion to the Binary Search Tree.

The third line contains a single integer X ($1 \leq X \leq 10^9$) — the value of the node whose subtree size you must calculate. It is guaranteed that X occurs among $a_1, a_2, \ldots, a_N$.

Output format

Print the size of the subtree of the given node.

Examples

Input

```
7
4 2 6 1 3 5 7
4
```

Output

```
7
```

Submit a solution for C-111632. Christmas Gifts

Time limit:	1 s
Real time limit:	5 s
Memory limit:	256M

Problem C: 111632. Christmas Gifts

Christmas is coming! Everyone is preparing gifts for their families. The Damir's family is also preparing for this. The Damir's family has many children, so the parents decided to buy n various gifts. Parents decided to number these gifts - i-th as a_i. They hang them on Christmas tree in socks, following the form binary search tree. In particular, they insert i-th gift with the value a_i following the rules of binary search tree.

As you know, Damir is the smallest among the whole family. Therefore, parents for the holiday allowed him to pick up his gift first. Damir knew that his gift has number k, but he mistakenly assumed that all the gifts below his gift were also intended for him. Now, parents are confused and want to find out what gifts Damir wants to grab for himself.

Input format

The first line contains one integer n ($1 \le n \le 10^3$) — the number of gifts.

The second line contains n pairwise distinct integers a_i ($1 \le a_i \le 10^3$) — the numbers of the gifts in the order they are hung.

The third line contains one integer k ($1 \le k \le 10^3$). It is guaranteed that the array contains the number k.

Output format

Print the numbers of the gifts of the subtree of k in pre-order: first the number of the current node, then the whole left subtree, then the whole right subtree. Print all the numbers in one line, separated by single spaces.

Examples

Input

```
5
4 2 7 1 3
2
```

Output

```
2 1 3
```

Examples

Input

```
5
4 2 7 1 3
2
```

Output

```
2 1 3
```

Notes

\linebreak

```
Given the tree:
        4
       / \
      2   7
     / \
    1   3

And the value to search: 2
```

You should return this subtree:

```
      2
     / \
    1   3
```

Submit a solution for D-105816. Aureole

Time limit:	1 s
Real time limit:	5 s
Memory limit:	256M

Problem D: 105816. Aureole

You are given a permutation of size n. Create an empty BST, and insert into BST values $p_1, p_2, \ldots, p_n$ in this order. You need to find how many levels are there and the sum of values for each level.

Input format

In the first line there is a single integer $1 \leq n \leq 5000$ size of permutation. Second line contains n distinct numbers from 1 to n - the permutation.

Output format

In the first line output k — the number of levels in the resulting BST. In the second line output k integers — the sum of values on each level, in order of increasing level, starting from level 0 (the root).

Examples

Input

```
1
1
```

Output

```
1
1
```

Input

```
5
4 3 5 1 2
```

Output

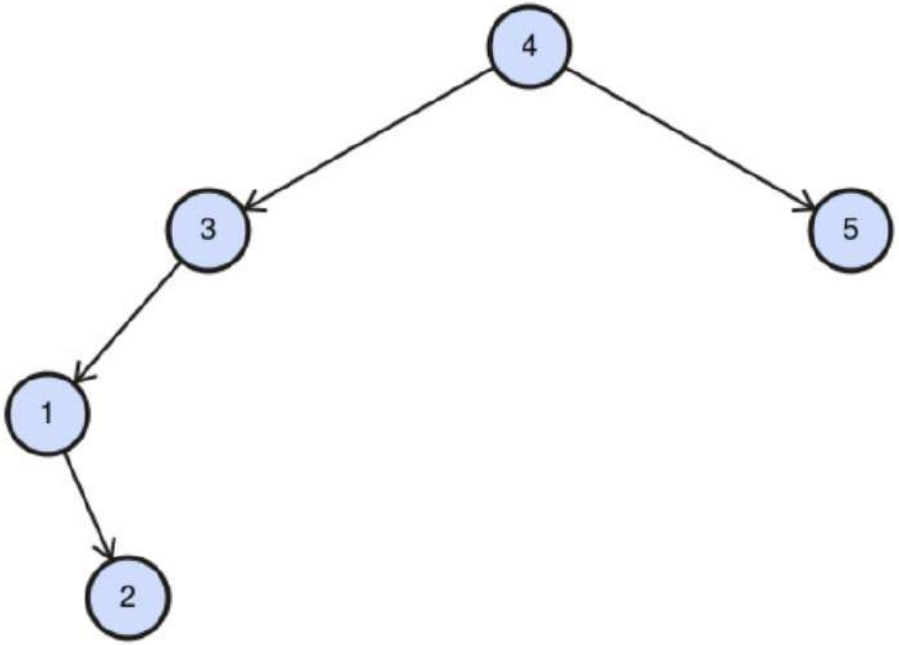
```
4
4 8 1 2
```

Notes

Level of vertex is defined as:

Level of root is 0, and level of each non-root vertex is (level of it's parent) + 1.

In second testcase, BST looks like this.



There are 4 levels, and sum for each level is 4, 3 + 5, 1, 2.

Problem E: 52477. Width

Given a binary tree, write a program to get the width of the given tree.

The level of a node is the number of vertices on the path from this node to the root. The width of a level h is the number of vertices with level h. The width of a tree is the maximal width over the levels.

Vertex number 1 always will be root.

Input format

The first line contains an integer n ($1 \le n \le 10^3$) – the number of vertices.

Each of the next $n-1$ lines contains 3 integers x, y, z ($1 \le x, y \le n, 0 \le z \le 1$) – vertex y is a son of vertex x; if $z = 0$ it is the left son, if $z = 1$ it is the right son. Vertex 1 is never a son. Every vertex except vertex 1 is a son of exactly one vertex, a vertex has at most one left and at most one right son, and the edges form a tree rooted at vertex 1. For $n = 1$ there are no such lines.

The $n-1$ lines are given in arbitrary order: a son may be described before its parent.

Output format

Print one integer maximum width.

Examples

Input

```
6
1 2 1
1 3 0
3 5 0
3 6 1
2 4 1
```

Output

```
3
```

Input

```
4
1 2 0
2 3 0
2 4 1
```

Output

```
2
```

Notes

In the first sample the widest level is the third one, it holds 3 vertices ($5, 6, 4$).

In the second sample the widest level is the third one as well, it holds 2 vertices ($3, 4$).

Problem F: 106735. Triangle Binary Search Tree

You are given N integers in order of their insertion to Binary Search Tree. You draw a set of horizontal lines, one through all the nodes of the same depth (the depth of the root is 0, and the depth of any other node is the depth of its parent plus one). After that you can see triangles with nodes instead of vertices and edges instead of sides. Your task is to calculate the number of the smallest triangles.

Input format

The first line consists of an integer N - number of nodes in Binary Search Tree ($1 \leqslant N \leqslant 10000$).

The second line contains N integers a_i - value of each node in Binary Search Tree in order of their insertion ($1 \leqslant a_i \leqslant N$).

It is guaranteed that there are no duplicates.

Output format

Print the number of mini-triangles in resulting Binary Search Tree.

Examples

Input

```
3
2 3 1
```

Output

```
1
```

Input

```
3
1 2 3
```

Output

```
0
```

Input

```
16
13 9 3 7 6 16 1 11 12 10 4 2 14 5 8 15
```

Output

```
5
```

Examples

Input

```
3
2 3 1
```

Output

```
1
```

Input

```
3
1 2 3
```

Output

```
0
```

Input

```
16
13 9 3 7 6 16 1 11 12 10 4 2 14 5 8 15
```

Output

```
5
```

Notes

A smallest triangle is formed by a node together with its left and its right son, so a node contributes a triangle exactly when both of its sons are present.

In the first sample the insertion order $2, 3, 1$ makes 2 the root with sons 1 and 3, so there is exactly 1 triangle.

In the second sample the values $1, 2, 3$ are inserted in increasing order, so the tree is a chain and no node has two sons — the answer is 0.

In the third sample the resulting tree contains 5 such triangles.

Problem G: 197831. Killua and Hunter exam

While Gon is surviving on the Greed Island, Killua, after the first unsuccessful attempt to pass the hunter exam, decides to test himself again. This time one of his tasks is to find the maximum distance between any two vertices in a binary search tree. Since Killua is pretty bad at algorithms, he asks for your help.

Input format

In the first line you will be given single number N $(1 \leq N \leq 20000)$. Next line consists of N numbers, where a_i $(1 \leq a_i \leq 10^9)$ represents the i-th number inserted to a binary search tree.

If a_i is already present in the tree, it is not inserted again — the tree never contains two nodes with equal values.

Subtasks

1. (30%) $N \leq 100$.
2. (30%) $N \leq 1000$
3. (40%) No additional constraints.

Every test is scored separately; the percentages above show the total weight of each group of tests.

Output format

Print one single number — the maximum distance between any two vertices of the binary search tree. The distance between two vertices is the number of vertices on the path between them, so for a tree of one vertex the answer is 1.

Examples

Input

```
9
11 5 3 2 1 7 9 8 13
```

Output

```
7
```

Input

```
5
1 2 4 3 5
```

Output

```
4
```

Input

```
7
4 2 6 5 1 3 7
```

Examples

Input

```
9
11 5 3 2 1 7 9 8 13
```

Output

```
7
```

Input

```
5
1 2 4 3 5
```

Output

```
4
```

Input

```
7
4 2 6 5 1 3 7
```

Output

```
5
```

Notes

In the first test, the answer is the distance between nodes 1 and 8.

Submit a solution for H-111743. Greater Sum Tree

Time limit:	1 s
Real time limit:	5 s
Memory limit:	256M

Problem H: 111743. Greater Sum Tree

Given the root of a binary search tree with distinct keys. Replace the key of each node with the sum of the keys over the nodes that has greater than or equal key. Print new keys in increasing order.

As a reminder, a binary search tree is a tree that satisfies these constraints:

- The left subtree of a node contains only nodes with keys less than the node's key.
- The right subtree of a node contains only nodes with keys greater than the node's key.
- Both the left and right subtrees must also be binary search trees.

Input format

The first line contains a single integer n $(1 \leq n \leq 100)$ — the number of nodes.

The second line contains n pairwise distinct integers a_i $(0 \leq a_i \leq 1000)$ — the keys, inserted one by one, in this order, into an initially empty binary search tree.

Output format

In a single line print n integers — the new keys in increasing order, separated by single spaces.

Examples

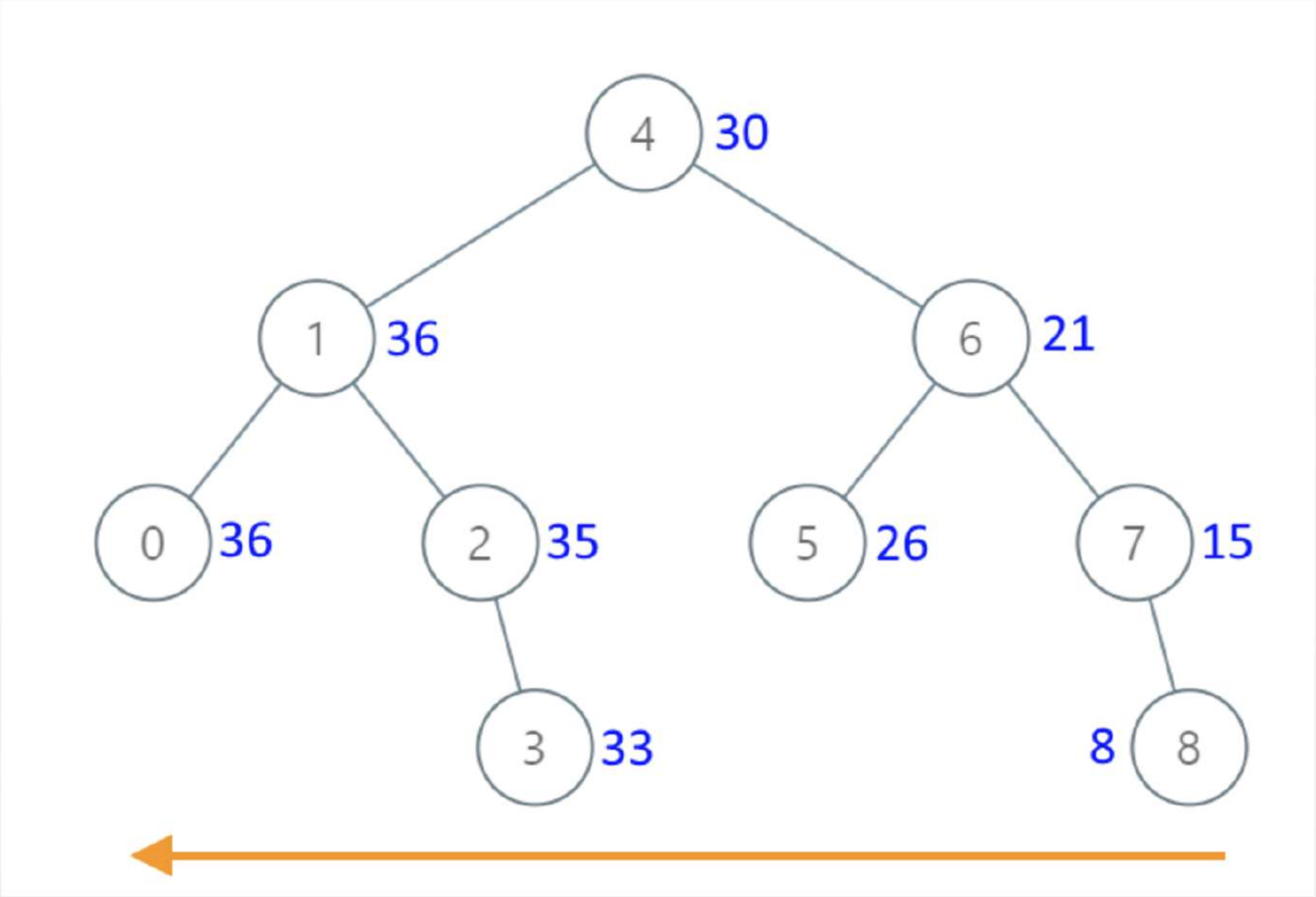
Input

```
9
4 1 6 0 2 3 5 7 8
```

Output

```
8 15 21 26 30 33 35 36 36
```

Notes



NOTE: Solve with BST!

Submit a solution for I-More One Night

Time limit:	1 s
Real time limit:	5 s
Memory limit:	256M

Problem I: More One Night

You are given a permutation of size n. Create an empty BST, and insert the values $p_1, p_2, \ldots, p_n$ into it in this order, one by one, using the usual binary search tree insertion (no balancing). Find the number of leaves in the resulting BST.

Input format

The first line contains a single integer n ($1 \le n \le 5000$) — the size of the permutation.

The second line contains n distinct integers from 1 to n — the permutation $p_1, p_2, \ldots, p_n$.

Output format

Output one integer — the number of leaves in the BST after all n insertions.

Examples

Input

```
1
1
```

Output

```
1
```

Input

```
5
4 3 5 1 2
```

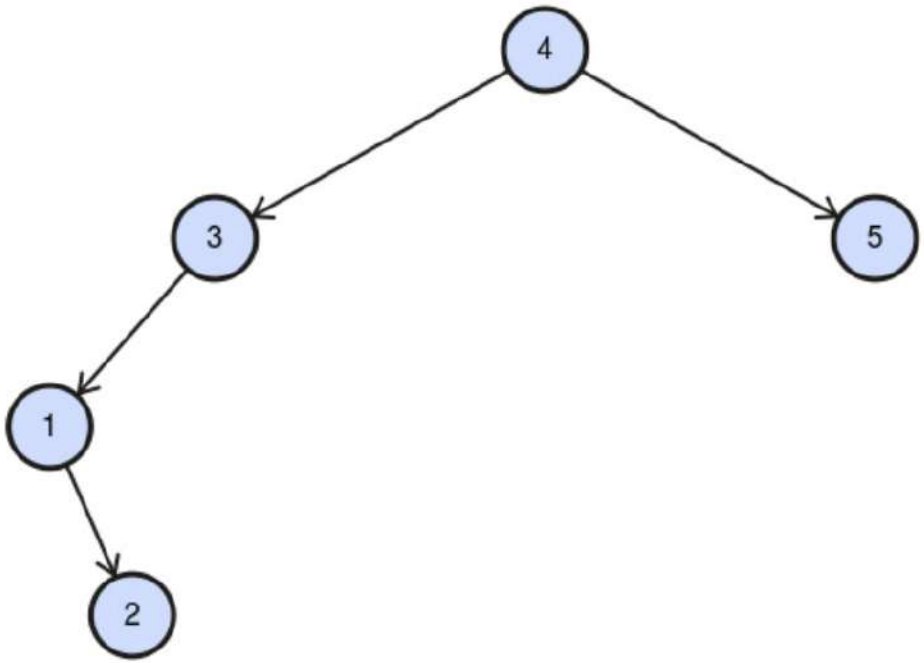
Output

```
2
```

Notes

A vertex is called a leaf if it has no children.

In the second sample the BST looks like this.



The answer is 2 (vertices 2 and 5).

Submit a solution for J-K-th element in Binary Search Tree

Time limit:	1 s
Real time limit:	5 s
Memory limit:	256M

Problem J: K-th element in Binary Search Tree

You are given N integers. Insert them one by one, in the given order, into an initially empty binary search tree, and then print the K-th smallest element of the tree.

For example, if the values are $\{20, 8, 22, 4, 12, 10, 14\}$ and $K = 3$, then the in-order traversal of the tree visits them in sorted order $\{4, 8, 10, 12, 14, 20, 22\}$, so the answer is 10; for $K = 5$ the answer is 14.

Input format

The first line contains two integers N and K ($1 \le N \le 10^5, 1 \le K \le 10^5$).

The second line contains N pairwise distinct integers a_i ($1 \le a_i \le 10^5$) — the values inserted into the tree, in this order.

Output format

Print the K-th smallest element of the tree.

If $K > N$, print -1.

Examples

Input

```
7 3
20 8 22 4 12 10 14
```

Output

```
10
```

Input

```
7 5
20 8 22 4 12 10 14
```

Output

```
14
```